# Shan Yang

Research Engineer — RL Post-Training Infrastructure for Multimodal & Agentic Models

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## Professional Summary

- **Ten years building ML systems end-to-end** — from a production video understanding stack at Google DeepMind to RL post-training pipelines at Amazon and Adobe. I build the training stacks, reward environments, and rollout systems that turn research ideas into trained models
- Most recently built the full **GSPO+DAPO training stack** (verl on FSDP, vLLM rollouts, rule-based and LLM-judge verifiers) that took Qwen3-VL-8B-Thinking from **8.0 to 26.3** on a held-out olympiad benchmark in ∼**360 H200-GPU-hours** across three runs (**Physics-R1**, sole-author arXiv 2026)
- Shipped **0-to-1 systems at scale**: Amazon's multimodal video search (deployed to customers), a 10B+ parameter video-generation stack for Prime Video, and a TensorFlow video object detection system at DeepMind that underpinned ICCV 2021 (**700+** citations) and NeurIPS 2021 (**800+** citations) papers

## Technical Stack

**Post-training algorithms:** SFT, DPO, RLHF, GRPO, GSPO, DAPO; Physics-R1 settled on GSPO+DAPO at 8B, the multi-turn follow-up runs decoupled advantages over a two-turn episode

**Training & rollout infrastructure:** verl 0.6.1 on FSDP; vLLM for rollout serving; PyTorch, DeepSpeed; TensorFlow (2017–2021); distributed training at 10B+ params

**Reward systems:** rule-based verifiers, LLM judges, unit-consistency checks, component-level reward logging so it is visible which term is driving the gradient; in-rollout tool harnesses (stop-token interception, sandboxed execution, `<tool_result>` feedback)

**Data pipelines:** multi-stage audit and contamination screening (lexical, embedding, then judge) gating every training corpus; held-out benchmark construction; preference-data pipelines

## Engineering & Research Experience

**Adobe Foundry** — San Jose, CA

Staff Applied Scientist — March 2026 – Present

- Own **end-to-end post-training for multi-reference image generation** on enterprise foundation models: data curation and SFT pipeline to date, now extending to **RLVR/RLAIF**
- Sole owner of post-training for multi-reference product-detail-page image generation from tech packs for an enterprise customer; own the **evaluation protocols, graders, and preference-data pipelines** that gate customer-model quality

**Amazon (Prime Video & Search)** — Palo Alto & Sunnyvale, CA

Senior Applied Scientist Tech Lead — September 2021 – March 2026

- **Search (2021–2025):** Built Amazon's **large-scale multimodal video search stack** (video / text / image / structured fusion), deployed to customers; applied **RL post-training** that reduced hallucination by **28%**; tech-led VLAP (ECCV 2024) and ICAR (AAAI 2024)
- **Prime Video (2025–2026):** Led the GenAI Live Action Studio **video-generation infrastructure** — autoregressive diffusion and training infrastructure for **10B+ parameter** models; controllable generation via test-time training and teacher-diffusion-forcing

**Google DeepMind** — Mountain View, CA

Senior Research Software Engineer — June 2017 – September 2021

- Built and maintained a **video object detection training and inference system in TensorFlow** for the research team, owning training and serving (through MediaPipe); owned 2D/3D pose reconstruction models deployed on GCP at scale
- This infrastructure underpinned AI Choreographer (ICCV 2021, co-first author, public AIST++ dataset, **700+** citations) and Attention Bottlenecks for Multimodal Fusion (NeurIPS 2021, second author, **800+** citations)

## In Progress

**Physics-R2 — multi-turn RL**: Denser reward for multimodal reasoning over a two-turn episode, with a sandboxed tool harness invoked mid-rollout and reward shaping over *when and how* the tool is called. Under submission.

## Open-source Artifacts

**physics-r1-code** — training and evaluation infrastructure: audit pipeline, GSPO+DAPO RL recipe, reward and judge harness (GitHub)

**PhysCorp-A** — 6,432-record audited training corpus with contamination screening, cut from a 14,294-record pool (HF)

**PhysOlym-A** — 500-problem held-out olympiad benchmark, 99.8% novel-source, EN/ET bilingual subset (HF)

**AIST++** — 1,408-clip motion-music dataset for 3D dance generation (GitHub, Google / ICCV 2021)

## Selected Publications

*Research outputs from the systems above.*

**Physics-R1: An Audited Olympiad Corpus and Recipe for Visual Physics Reasoning**
***S Yang***
arXiv 2026 (2605.14040) — project / code / data
*Sole-author. Three-stage contamination audit (14,294 → 6,432 records), 500-problem held-out benchmark, composite reward system, and GSPO+DAPO recipe on verl/FSDP with vLLM rollouts; +18.3 pp on PhysOlym-A, +15.7 pp on PhysReason at 8B.*

**Beyond Copy-Paste: How Well Do Subject-Driven Video Models Understand Their Subjects?**
*Z Wang, K Deng, D Blackburn, L Lu, J Liang, Y Lou, M Bansal,* ***S Yang***
NeurIPS 2026

**Beyond Test-Time Training: Learning to Reason via Hardware-Efficient Optimal Control**
*P Wang,* ***S Yang****, X Wang, T Xiao, X Liu, C Yu, Y Lou, P Li, Z Wang, MC Lin, R Vidal*
ICML 2026
*Optimal-control formulation of LLM test-time reasoning (+35.8% Math-500, +10% AIME25).*

**VLAP: Efficient Video-Language Alignment via Frame Prompting and Distilling for Video Question Answering**
*X Wang, J Liang, CK Wang, K Deng, Y Lou, MC Lin,* ***S Yang***
ECCV 2024

**ICAR: Image-based Complementary Auto Reasoning**
*X Wang, A Liang, J Liang, M Lin, Y Lou,* ***S Yang***
AAAI 2024

**Attention Bottlenecks for Multimodal Fusion**
*A Nagrani,* ***S Yang****, A Arnab, A Jansen, C Schmid, C Sun*
NeurIPS 2021 — **800+ citations**

**AI Choreographer: Music Conditioned 3D Dance Generation with AIST++**
*R Li*,* ***S Yang*****, DA Ross, A Kanazawa*
ICCV 2021 — **700+ citations**

**Physics-Inspired Garment Recovery from a Single-View Image**
***S Yang****, Z Pan, T Amert, K Wang, L Yu, T Berg, MC Lin*
ACM ToG 2018 — **100+ citations**

**Modeling Context in Referring Expressions**
*L Yu, P Poirson,* ***S Yang****, A Berg, T Berg*
ECCV 2016 — **1600+ citations**

## Education

**University of North Carolina at Chapel Hill** — Chapel Hill, NC
Ph.D., Computer Science — 2012 – 2017

**Shanghai Jiao Tong University** — Shanghai, China
B.E., Software Engineering — 2008 – 2012